

Moment-to-Moment Affect During Film-Induced Somatic Anxiety

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Author Note

Draft Methods and Results for internal review. Preregistration: <https://osf.io/s2yvb>.
The confirmatory tests were run on the confirmatory sample from commit 63d042d (random seed 20261004); every number in this document is generated directly from those outputs.

Method

The study was preregistered on the Open Science Framework before any hypothesis test was run (<https://osf.io/s2yvb>). Hypotheses, segment edges, exclusion rules, tests, correction families and inference criteria are reported here as registered, and anything not in the registration is labelled exploratory. Decisions taken after registration are recorded, with dates, in the project's preregistration log and summarized at the end of the Results.

Participants

Participants were recruited on Prolific. Eligibility filters were fluent English, normal or corrected-to-normal vision, an approval rate of at least 95%, at least 20 previous submissions and a desktop computer; the study listing and information sheet also required a physical mouse or trackpad. Recruitment ran in cohorts. Each cohort was a separate Prolific study with a 50/50 quota on Prolific's sex screener, and participants from earlier cohorts were blocked from later ones. Cohort 4 was a single study whose places were increased in steps of 20 about 25 minutes apart, which limited the number of simultaneous video downloads. Participants were paid £4.80 for an estimated 30 minutes. The information sheet advised people with a strong fear of heights not to take part, and consent included the statement "I do not expect this to be seriously distressing for me." The study was conducted under the Aarhus University standard waiver for non-interventional online studies.

The pilot and cohort 1 form the exploratory sample, which was used to set the segment edges, the exclusion rules and the sample size. Cohorts 2–4 form the confirmatory sample, and all confirmatory tests use only these participants. The registered target was 120 usable confirmatory participants, with recruitment stopping at the end of the first cohort after which this number was reached. This sample size gives 80% power to detect correlations of about .25 (two-sided $\alpha = .05$); the within-participant effects in the exploratory data were far larger.

Stimulus

The film shows people climbing a skyscraper from a first-person view. It lasts 13 min 28 s and was presented without sound (H.264, 1280 × 720 pixels, 25 frames per second). The whole

file was downloaded before playback began; if the download did not finish within 5 minutes, the session ended and the participant was asked to return the study.

Procedure

The task ran in the browser (jsPsych 8 on JATOS). After giving consent, participants passed a device check (desktop computer, browser window of at least 1000×600 pixels), rated their everyday fear of heights and completed the first attention check. The study then switched to full screen. Participants read the instructions and practised the rating task: they tracked a moving target on the two-dimensional grid for 40 s (repeated once if the tracking error exceeded 20 units) and then placed six example emotional states on the grid.

Participants then completed the STICSA somatic scale, watched the film while rating continuously, and completed the STICSA somatic scale again immediately afterwards. The session ended with the post-film questions, the second attention check, a written description of the experience of at least 100 words and an optional feedback field. If participants switched tabs, left full screen or released the mouse pointer, the film paused until they returned.

Measures

Continuous Valence and Arousal

Throughout the film, participants moved a dot within a small affect grid in the corner of the video using the mouse (pointer locked). The horizontal position coded valence (0 = unpleasant, 50 = neutral, 100 = pleasant) and the vertical position coded arousal (0 = low, 100 = high activation). The position was sampled at 20 Hz and time-stamped against film playback time.

Somatic Anxiety

State somatic anxiety was measured with the somatic subscale of the State-Trait Inventory for Cognitive and Somatic Anxiety (STICSA; state form), immediately before and immediately after the film, with the instruction “Please indicate how you feel right now, at this moment.” The subscale has 11 items (items 1, 2, 6, 7, 8, 12, 14, 15, 18, 20 and 21), each answered from 1 (*not at all*) to 4 (*very much so*). The total ranges from 11 to 44 and was computed only when all items

were answered; the change score is post minus pre.

Single-Item Ratings

Before the film, participants answered “In everyday life, how afraid are you of heights?” from 1 (*not at all*) to 7 (*extremely*). After the film, they rated on 1–7 scales how tense or anxious they had felt overall while watching (“Overall, how tense or anxious did you feel while watching the film?”), how concerned they had been for the safety of the people in the film, and how strong their bodily sensations had been. They also reported whether they had seen the film before, whether they had watched attentively and which input device they had used.

Data Processing

For each participant, the 20-Hz ratings were resampled to 1 Hz of film time (808 values, 0–807 s) by linear interpolation over the samples recorded while the film was playing. The film was divided into four segments with edges fixed in advance: baseline (0–262 s), rise (262–470 s), plateau (470–692 s) and return (692–808 s). These edges had been estimated in the exploratory sample by piecewise-linear segmentation of the group-mean arousal series. A participant’s segment mean is the mean of their 1-Hz series within a segment, and the whole-film mean is the mean over all 808 values. Group-mean series are the mean across participants at each second.

Exclusions

Participants who completed the study were excluded if any of seven registered rules applied, all decided from data-quality logs without reference to the hypotheses: (a) failing both attention checks; (b) script-generated input events during the film; (c) a written description that was not typed (keystrokes fewer than 0.2 times the number of characters, for texts of at least 200 characters); (d) erratic input (more than 10% of mouse movements larger than 250 pixels); (e) disrupted viewing (more than 30 s of buffering or more than 60 s of interruptions during the film); (f) ratings recorded during fewer than 95% of the film’s seconds; or (g) having seen the film before.

Two sensitivity samples were also registered, and every confirmatory test was repeated in each. The first additionally excluded sparse raters (fewer than one rating bout per minute, where

inputs more than 2 s apart start a new bout, or a stretch of more than 180 s without input), participants who failed one of the two attention checks, and participants who reported not watching attentively. The second additionally excluded participants whose Prolific submissions had been rejected for their behaviour on the free-text page but who passed the registered exclusion rules.

Statistical Analysis

All tests were two-sided with $\alpha = .05$ after Holm correction within each family of hypotheses. A directional hypothesis was considered supported only if the effect was significant in the predicted direction.

H1. STICSA somatic totals after and before the film were compared with a paired t test. The effect size is Cohen's d_z (the mean of the differences divided by their standard deviation) with an exact 95% confidence interval from the noncentral t distribution. A Wilcoxon signed-rank test was run as a robustness check.

H2. Four within-participant contrasts of segment means were tested with paired t tests: arousal in the rise and on the plateau against baseline (both predicted higher), arousal in the return against the plateau (predicted lower) and valence on the plateau against baseline (predicted lower). Effect sizes are d_z with exact 95% confidence intervals; p values were Holm-corrected across the four contrasts.

H3. For arousal and for valence separately, the group-mean series of the confirmatory sample was correlated with the group-mean series of the exploratory sample (Pearson r over the 808 one-second values). Because neighbouring seconds are not independent, the 95% confidence interval was obtained with a circular block bootstrap over time (30-s blocks, 5,000 resamples, the same blocks drawn from both series). H3 was considered supported for a dimension if the lower bound of this interval exceeded .50.

H4–H6. The individual-difference hypotheses were tested with Spearman rank correlations across participants, chosen because the predictors are ordinal and the outcomes bounded. H4 related everyday fear of heights to plateau arousal (H4a, predicted positive), plateau

valence (H4b, negative) and the STICSA change (H4c, positive). H5 related the STICSA change to whole-film mean arousal (H5a, positive) and valence (H5b, negative). H6 related post-film overall anxiety to the STICSA change (H6a, positive) and to plateau arousal (H6b, positive). Each correlation has a percentile bootstrap 95% confidence interval (5,000 resamples of participants), and Holm correction was applied within H4 (three tests), H5 (two) and H6 (two). The trend lines in Figure 5 are least-squares fits drawn as a visual guide; they are not the tests.

Descriptive Analyses. Two registered descriptive analyses accompany the tests. The reliability of the group-mean series was estimated by splitting the participants randomly into two halves 1,000 times, correlating the two half-sample means and applying the Spearman–Brown correction. The segment edges were re-estimated in the confirmatory sample with the method used in the exploratory sample: an exact piecewise-linear segmentation of the group-mean arousal series into four straight-line segments (three breakpoints, each segment at least 30 s long, found by dynamic programming), with 95% intervals from 1,000 bootstrap resamples of participants.

Exploratory Analyses. Two analyses were not registered as tests and are reported as exploratory. The first describes how participants occupied the affect grid in each segment. The second examines age and sex, which the registration lists among the planned exploratory analyses. Sex was taken from participants' Prolific profiles (female or male); Prolific withholds age and sex for rejected submissions, so these analyses include only participants with both. The STICSA change was regressed on sex and age. For the segment means of each rating, a mixed model with a random intercept per participant included segment (sum-coded), sex, age and the segment $\times$ sex and segment $\times$ age interactions; each interaction was tested with a likelihood-ratio test (3 degrees of freedom) against the model without it. As a check, the same model was fitted by least squares with standard errors clustered by participant and the interactions tested with Wald F tests. The age and sex analyses, including the rule that the mixed model is the main result, were specified and tried out on the exploratory sample before they were run on the confirmatory sample. There is no registered correction family for these analyses, so p values are uncorrected and the estimates with their confidence intervals should carry the weight.

Missing Data and Software. STICSA items were required, so all participants had complete totals, and no rating series had gaps after resampling; every test therefore used all participants in the sample. Analyses were run in Python 3.13.7 with NumPy 2.5.3, pandas 3.0.6, SciPy 1.18.1, Matplotlib 3.11.2 and, for the mixed models, statsmodels 0.15.0, using the random seed 20261004 for all resampling. Each analysis run saved a record of the code version, input-file checksums, settings and package versions.

Results

Sample

Of the 130 participants in the confirmatory sample who completed the study, 6 were excluded under the registered rules: 3 for disrupted viewing, 2 because their written description was not typed and 1 for erratic mouse input. No participant met any of the other four rules. The primary sample therefore comprised $N = 124$ participants (19, 20 and 85 from cohorts 2, 3 and 4). Table 1 describes the confirmatory and exploratory samples. Participants in the primary sample were on average 34.2 years old ($SD = 12.0$, range 18–76); 59 reported their sex as female and 62 as male, and Prolific withheld age and sex for the 3 participants whose submissions had been rejected. Most rated with a mouse (80); 43 used a trackpad.

The stricter-engagement sensitivity sample comprised $n = 96$ participants after removing 24 sparse raters, 5 participants who failed one attention check and 3 who reported not watching attentively (these groups overlap). Removing the 4 participants in the rejection set left $n = 120$. The reference curve for H3 was the group mean of the 21 usable exploratory participants (of 23 who completed the study).

Descriptive Statistics

Table 2 summarizes the measures used in the confirmatory tests. Somatic anxiety was low before the film: 45 participants (36%) had the lowest possible STICSA somatic score of 11. Everyday fear of heights covered the whole scale, but the top was thinly populated; only 20 participants rated their fear at 6 or 7, and 2 chose 7. This fits the recruitment material, which advised people with a strong fear of heights not to take part. Participants reported feeling clearly

Table 1*Sample Characteristics*

Characteristic	Exploratory, usable	Confirmatory, all completers	Confirmatory, primary sample
<i>n</i>	21	130	124
Age, years, <i>M</i> (<i>SD</i>)	35.0 (12.6)	34.1 (11.9)	34.2 (12.0)
Range	19–60	18–76	18–76
Sex, <i>n</i> (%)			
Female	13 (62)	62 (48)	59 (48)
Male	8 (38)	64 (49)	62 (50)
Not available ^a	0 (0)	4 (3)	3 (2)
Input device, <i>n</i> (%)			
Mouse	15 (71)	86 (66)	80 (65)
Trackpad	6 (29)	43 (33)	43 (35)
Other	0 (0)	1 (1)	1 (1)
Everyday fear of heights (1–7), <i>M</i> (<i>SD</i>)	3.52 (1.72)	3.72 (1.73)	3.77 (1.73)
STICSA somatic, pre-film (11–44), <i>M</i> (<i>SD</i>)	12.24 (1.81)	13.72 (3.51)	13.76 (3.56)
Failed one attention check, <i>n</i> (%)	0 (0)	5 (4)	5 (4)
Excluded under registered rules, <i>n</i>	0	6	0

Note. Exploratory sample: pilot and cohort 1, usable under the registered rules (reference curve for H3). Confirmatory sample: cohorts 2–4. Age and sex are Prolific profile data; input device is self-reported. STICSA = State-Trait Inventory for Cognitive and Somatic Anxiety, state somatic subscale. ^aProlific withholds demographic data for rejected submissions.

tense or anxious while watching ($M = 5.18$ on a 1–7 scale) and were very concerned for the safety of the people in the film ($Mdn = 7$).

The continuous ratings traced the expected course (Figure 1). Mean arousal was moderate during the baseline segment, climbed through the rise, stayed high across the plateau and fell again during the return. Valence moved in the opposite direction, from close to the neutral midpoint at baseline to clearly unpleasant on the plateau, and back to neutral by the end of the film.

Confirmatory Analyses

Table 3 lists all preregistered tests for the primary sample. Ten of the 12 directional predictions were supported, as were both replication criteria.

Table 2*Descriptive Statistics, Primary Confirmatory Sample*

Measure	Scale	<i>M</i>	<i>SD</i>	<i>Mdn</i>	Range
<i>Before the film</i>					
Everyday fear of heights	1–7	3.77	1.73	4.00	1–7
STICSA somatic, pre-film	11–44	13.76	3.56	13.00	11–27
<i>After the film</i>					
STICSA somatic, post-film	11–44	20.55	7.29	18.00	11–44
STICSA somatic change (post – pre)		6.79	6.60	5.00	–6–32
Overall anxiety while watching	1–7	5.18	1.33	5.00	1–7
Concern for the climbers' safety	1–7	6.13	1.30	7.00	1–7
Intensity of bodily sensations	1–7	4.64	1.60	5.00	1–7
<i>Continuous arousal</i>					
Baseline (0:00–4:22)	0–100	43.13	15.80	43.76	5.9–76.4
Rise (4:22–7:50)	0–100	67.14	14.79	68.91	22.5–96.1
Plateau (7:50–11:32)	0–100	80.91	15.01	83.07	24.2–100.0
Return (11:32–13:28)	0–100	53.89	19.42	54.76	13.3–96.9
Whole film	0–100	61.23	10.99	60.58	21.2–86.7
<i>Continuous valence</i>					
Baseline (0:00–4:22)	0–100	50.58	14.55	51.96	12.6–85.2
Rise (4:22–7:50)	0–100	39.58	19.06	38.52	0.2–83.3
Plateau (7:50–11:32)	0–100	29.13	23.10	23.89	0.0–100.0
Return (11:32–13:28)	0–100	50.84	20.44	51.67	0.6–97.9
Whole film	0–100	41.89	14.67	41.91	4.6–86.5

Note. $N = 124$. Range is the observed minimum–maximum. Continuous ratings are participant means of the 1-Hz series within each registered segment (film time in min:s); valence 0 = unpleasant, 50 = neutral, 100 = pleasant; arousal 0 = low, 100 = high. STICSA = State-Trait Inventory for Cognitive and Somatic Anxiety, state somatic subscale.

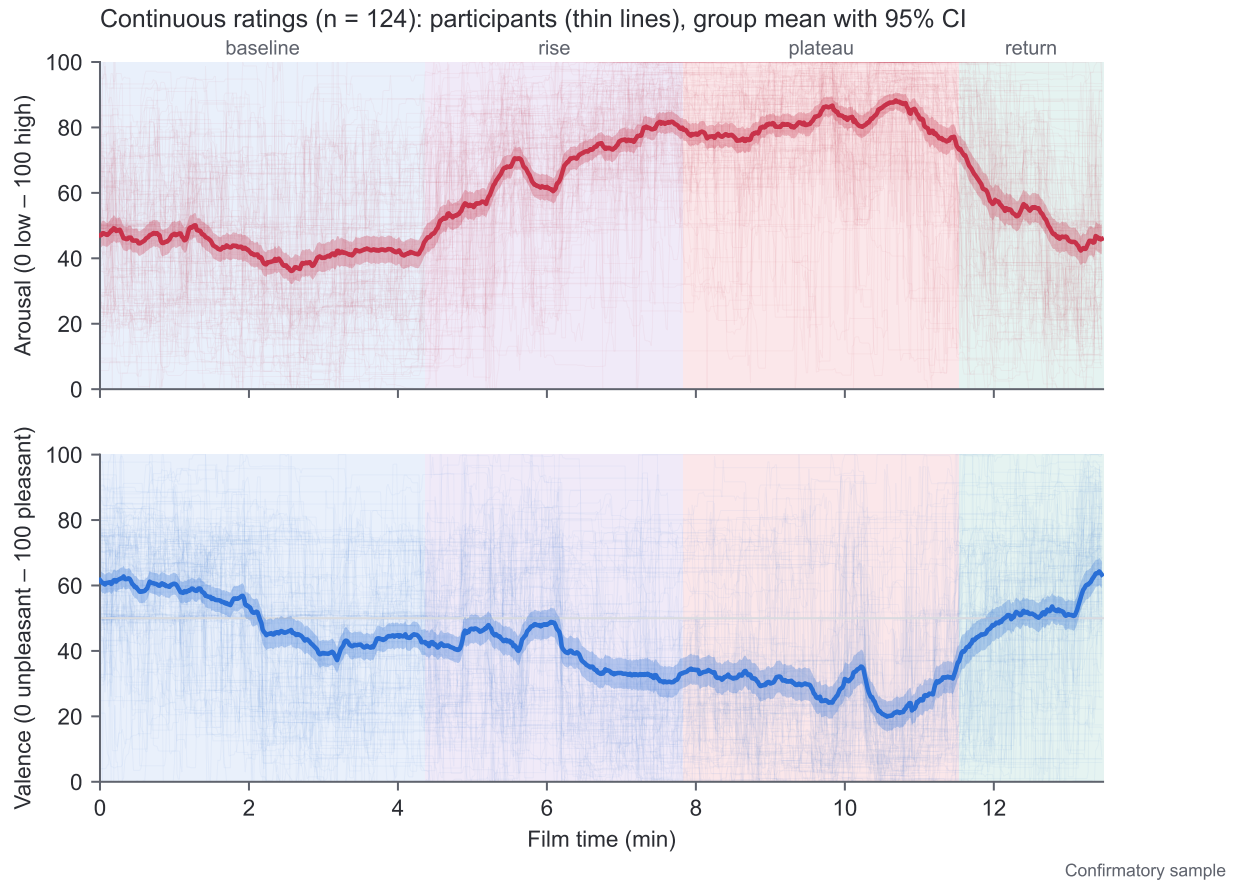
Somatic Anxiety Induction (H1)

STICSA somatic scores were higher immediately after the film than immediately before it, by 6.79 points on average (pre: $M = 13.76$, $SD = 3.56$; post: $M = 20.55$, $SD = 7.29$), $t(123) = 11.46$, $p < .001$, $d_z = 1.03$, 95% CI [0.81, 1.25]. Scores rose for 109 participants, did not change for 7 and fell for 8. Because many pre-film scores sat at the floor of the scale, we also ran the registered Wilcoxon signed-rank test, which gave the same result ($T = 197$, $p < .001$). The increase was spread across all eleven items rather than driven by one or two symptoms (Figure 2); a racing heart and muscle tension were the sensations most often endorsed after the film. H1 was

Table 3*Preregistered Confirmatory Tests, Primary Sample*

Hypothesis	Prediction	Test statistic	Effect size [95% CI]	p	p_{Holm}	Supported
H1	STICSA somatic: post > pre	$t(123) = 11.46$	$d_z = 1.03 [0.81, 1.25]$	< .001	–	Yes
H2a	Arousal: rise > baseline	$t(123) = 14.67$	$d_z = 1.32 [1.07, 1.56]$	< .001	< .001	Yes
H2b	Arousal: plateau > baseline	$t(123) = 19.77$	$d_z = 1.78 [1.49, 2.06]$	< .001	< .001	Yes
H2c	Arousal: return < plateau	$t(123) = -14.70$	$d_z = -1.32 [-1.56, -1.08]$	< .001	< .001	Yes
H2d	Valence: plateau < baseline	$t(123) = -9.85$	$d_z = -0.88 [-1.09, -0.68]$	< .001	< .001	Yes
H3 arousal	Arousal curve replicates	–	$r = .993 [.990, .996]$	–	–	Yes
H3 valence	Valence curve replicates	–	$r = .937 [.893, .966]$	–	–	Yes
H4a	Fear of heights, plateau arousal (+)	–	$\rho = .08 [-.09, .26]$	.355	.355	No
H4b	Fear of heights, plateau valence (–)	–	$\rho = -.32 [-.46, -.16]$	< .001	.001	Yes
H4c	Fear of heights, STICSA change (+)	–	$\rho = .18 [.00, .35]$	.049	.098	No
H5a	STICSA change, whole-film arousal (+)	–	$\rho = .24 [.05, .41]$	.008	.008	Yes
H5b	STICSA change, whole-film valence (–)	–	$\rho = -.38 [-.51, -.23]$	< .001	< .001	Yes
H6a	Overall anxiety, STICSA change (+)	–	$\rho = .56 [.44, .67]$	< .001	< .001	Yes
H6b	Overall anxiety, plateau arousal (+)	–	$\rho = .34 [.15, .50]$	< .001	< .001	Yes

Note. $N = 124$. H1 and H2: paired t tests with d_z and exact 95% CIs (noncentral t). H3: Pearson r between the confirmatory and exploratory group-mean series (808 one-second values), 95% CI from a circular block bootstrap (30-s blocks, 5,000 resamples); supported if the lower bound exceeds .50. H4–H6: Spearman ρ with percentile bootstrap 95% CIs (5,000 resamples). Holm correction within H2 (four tests), H4 (three), H5 (two) and H6 (two); H1 is a single test. All tests two-sided; a directional hypothesis counts as supported only if significant in the predicted direction.

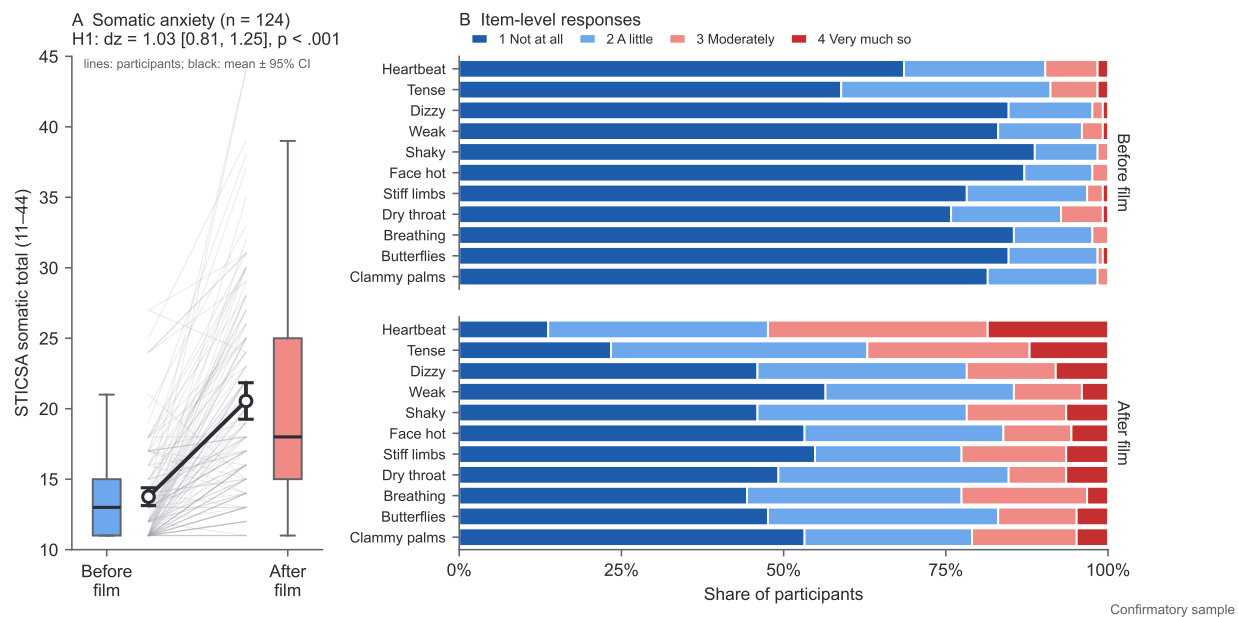
Figure 1*Continuous Arousal and Valence Ratings Over the Film*

Note. Thin lines show individual participants; the thick line is the group mean with its 95% confidence interval ($N = 124$). Shading marks the four preregistered segments (baseline, rise, plateau, return).

supported.

Affective Time Course (H2)

All four segment contrasts went in the predicted direction and remained significant after Holm correction (Figure 3). Compared with the baseline, arousal was higher during the rise, by 24.01 points on the 0–100 scale, $t(123) = 14.67$, $p_{\text{Holm}} < .001$, $d_z = 1.32$, 95% CI [1.07, 1.56] (H2a), and higher still during the plateau, by 37.78 points, $t(123) = 19.77$, $p_{\text{Holm}} < .001$, $d_z = 1.78$, 95% CI [1.49, 2.06] (H2b). From the plateau to the return, arousal fell by 27.02 points, $t(123) = -14.70$, $p_{\text{Holm}} < .001$, $d_z = -1.32$, 95% CI [-1.56, -1.08] (H2c). Valence was lower on the plateau

Figure 2*Somatic Anxiety Before and After the Film*

Note. (A) STICSA somatic totals. Grey lines connect each participant's scores; boxes show medians and interquartile ranges; the black line shows means with 95% confidence intervals. (B) Share of participants giving each answer to the eleven somatic items before (top) and after (bottom) the film.

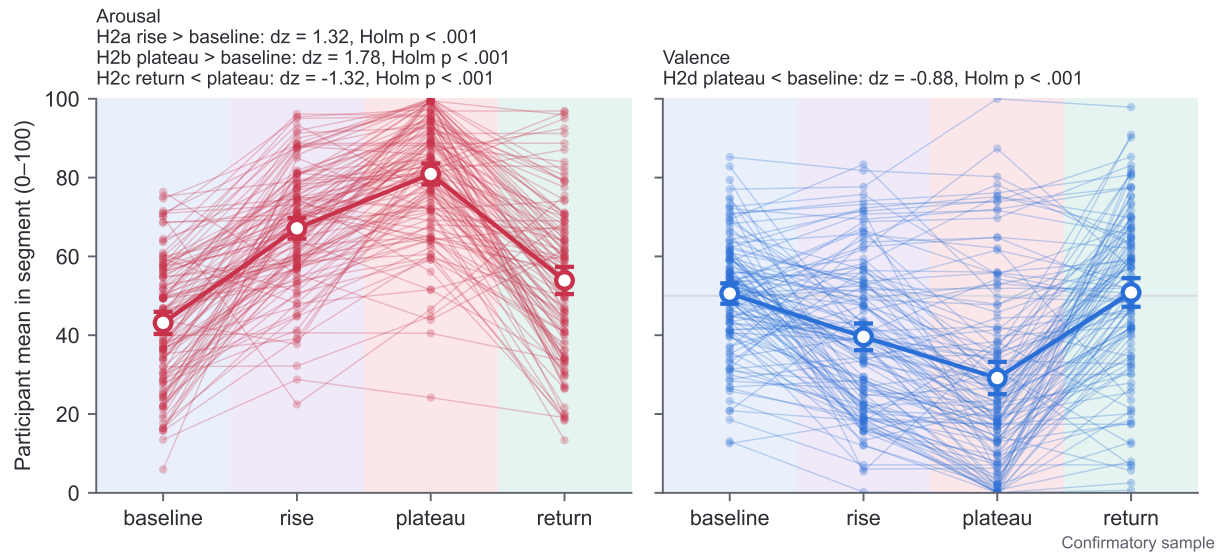
than at baseline, by 21.44 points, $t(123) = -9.85$, $p_{\text{Holm}} < .001$, $d_z = -0.88$, 95% CI [-1.09, -0.68] (H2d). The arousal effects were larger than the valence effect, partly because valence ratings varied more between participants on the plateau ($SD = 23.10$, against 15.01 for arousal).

Replication of the Group Time Course (H3)

The group-mean time series of the confirmatory sample closely followed that of the independent exploratory sample (Figure 4). For arousal, $r = .993$, 95% CI [.990, .996], and for valence, $r = .937$, 95% CI [.893, .966]. In both cases the lower bound of the confidence interval lay well above the preregistered criterion of .50, so H3 was supported for both dimensions. The two arousal curves had almost the same shape but differed somewhat in level: the confirmatory sample rated arousal a little higher during the baseline and the final minutes, and a little lower on the plateau.

Figure 3*Mean Ratings in the Four Preregistered Segments*

Segment means (n = 124): participants (thin lines), mean $\pm$ 95% CI



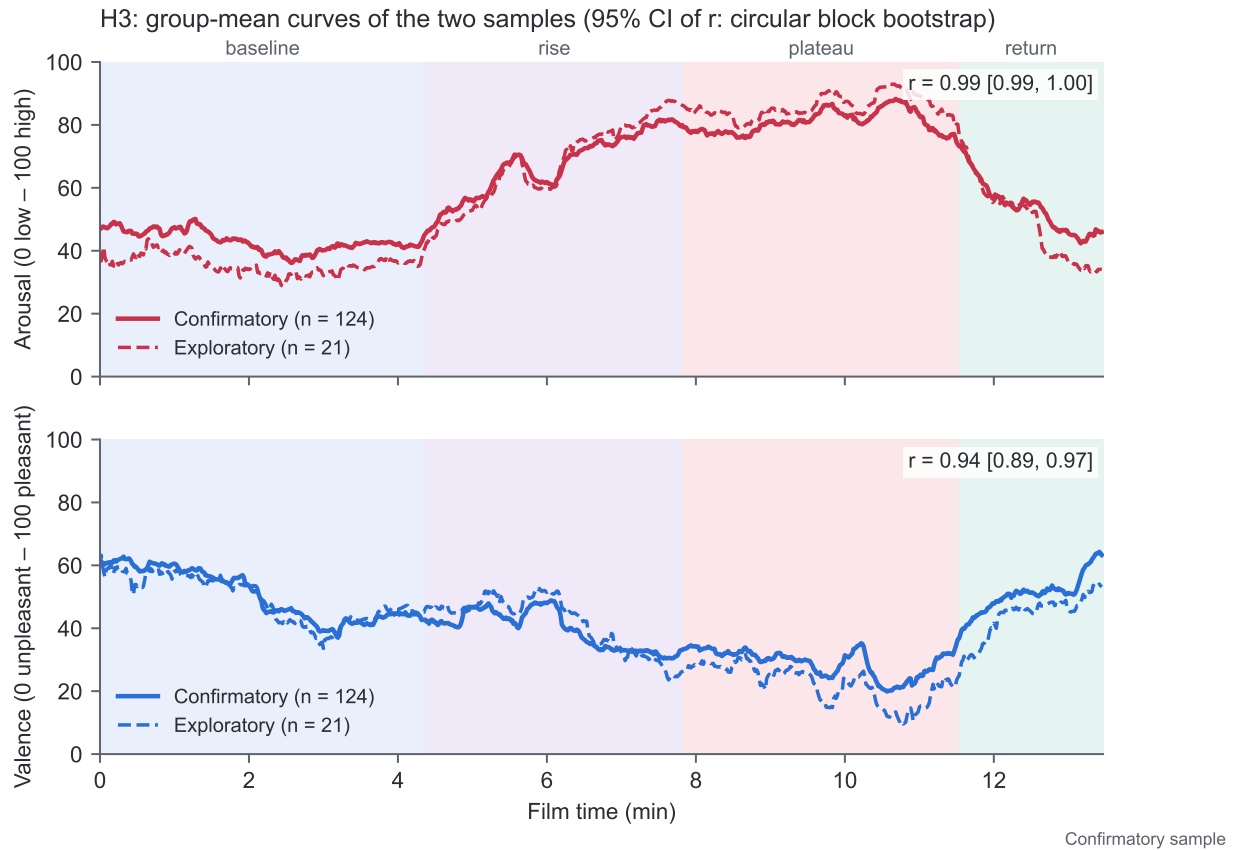
Note. Each thin line is one participant's segment means; the thick line shows group means with 95% confidence intervals. Panel headings give the H2 contrasts with d_z and Holm-adjusted p values.

Individual Differences (H4–H6)

Everyday fear of heights was related to how unpleasant the plateau felt, but not to how arousing it was or to the change in somatic anxiety (Figure 5). Participants with a greater fear of heights rated valence on the plateau as more negative, $\rho = -.32$, 95% CI $[-.46, -.16]$, $p < .001$, $p_{\text{Holm}} = .001$ (H4b). The predicted positive correlations with plateau arousal, $\rho = .08$, 95% CI $[-.09, .26]$, $p = .355$, $p_{\text{Holm}} = .355$ (H4a), and with the STICSA change, $\rho = .18$, 95% CI $[-.00, .35]$, $p = .049$, $p_{\text{Holm}} = .098$ (H4c), were not supported; the latter was significant only before correction.

The change in somatic anxiety was related to the continuous ratings as predicted. Participants whose STICSA scores rose more gave higher arousal ratings over the whole film, $\rho = .24$, 95% CI $[-.05, .41]$, $p = .008$, $p_{\text{Holm}} = .008$ (H5a), and lower valence ratings, $\rho = -.38$, 95% CI $[-.51, -.23]$, $p < .001$, $p_{\text{Holm}} < .001$ (H5b).

Overall anxiety reported after the film converged with both other measures. It correlated

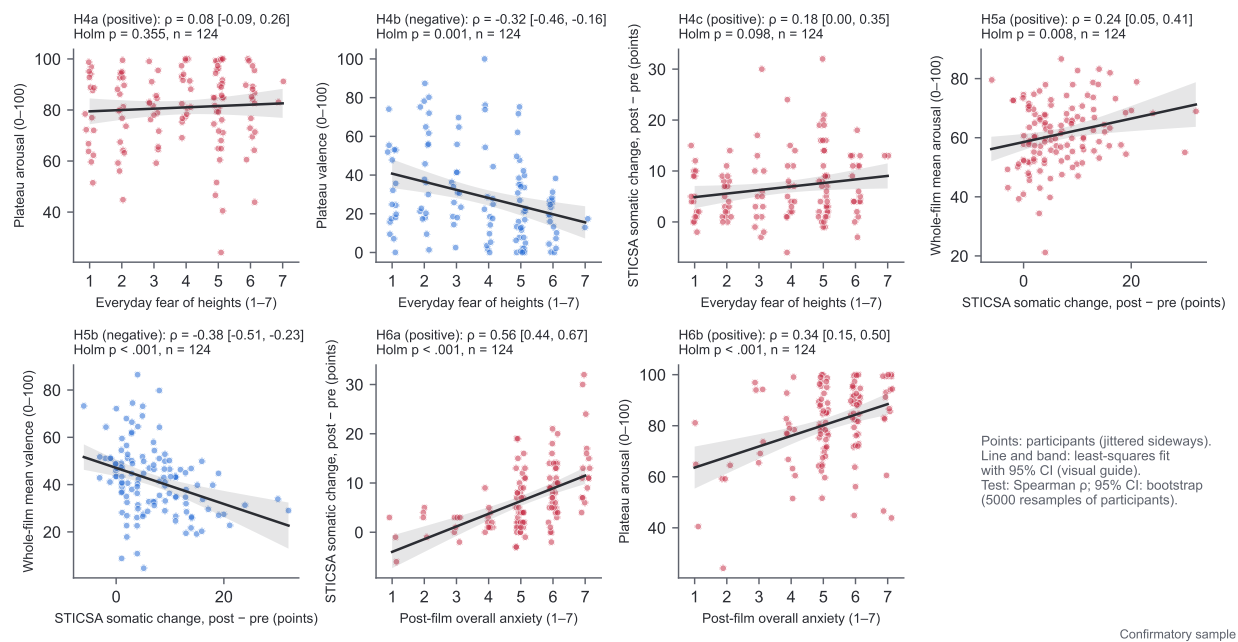
Figure 4*Group-Mean Time Series in the Confirmatory and Exploratory Samples*

Note. Solid lines: confirmatory sample ($n = 124$); dashed lines: exploratory sample ($n = 21$). Pearson r is computed over the 808 one-second values; the 95% confidence interval comes from a circular block bootstrap with 30-s blocks.

with the STICSA change, $\rho = .56$, 95% CI [.44, .67], $p < .001$, $p_{\text{Holm}} < .001$ (H6a), and with plateau arousal, $\rho = .34$, 95% CI [.15, .50], $p < .001$, $p_{\text{Holm}} < .001$ (H6b). Both H5 and H6 were supported.

Sensitivity Analyses

Repeating every test in the two sensitivity samples did not change any conclusion (Table 4). The same ten directional predictions and both replication criteria were supported in each sample, and H4a and H4c were not supported in either. Effect sizes stayed close to those in

Figure 5*Individual Differences Tested in H4–H6*

Note. Points are participants (shifted sideways slightly so that equal scores remain visible). The black line and grey band show a least-squares fit with its 95% confidence interval, drawn as a visual guide; the tests are Spearman correlations. Panel headings give Spearman ρ with bootstrap 95% confidence intervals and Holm-adjusted p values.

the primary sample; the segment contrasts for arousal were somewhat larger when sparse and less engaged raters were removed.

Preregistered Descriptive Analyses

The group-mean time series were highly reliable. Split-half reliability (1,000 random splits, Spearman–Brown corrected) was .992 for arousal and .979 for valence.

Re-estimating the segment edges in the confirmatory sample with the same piecewise-linear method placed the first edge at 262 s (participant-bootstrap 95% interval [133, 271]), identical to the registered value of 262 s, and the second at 467 s [317, 475], against 470 s registered. The third edge, the start of the return, came out at 660 s [631, 690], about half a minute earlier than the registered 692 s, which lies just outside the interval. The first two intervals are

Table 4*Preregistered Tests Repeated in the Sensitivity Samples*

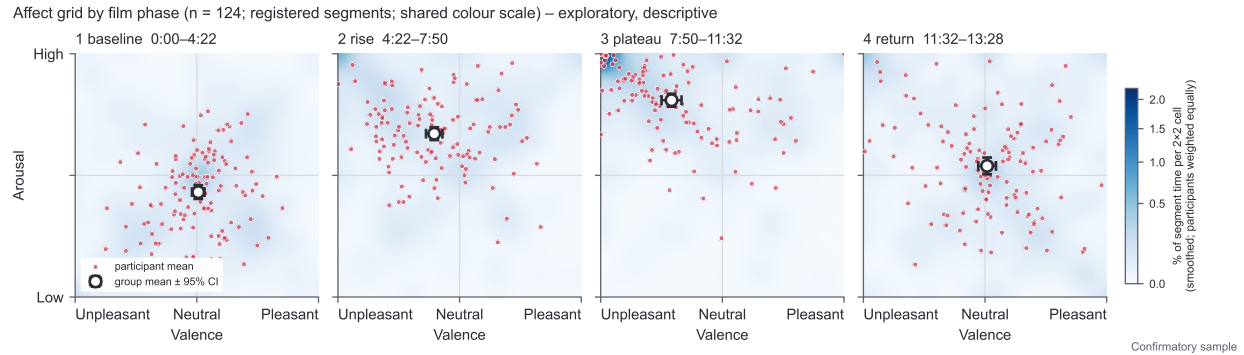
Hypothesis	Effect	Primary ($n = 124$)	Stricter engagement ($n = 96$)	Rejection set removed ($n = 120$)
H1	d_z	1.03 ($p < .001$)	1.09 ($p < .001$)	1.02 ($p < .001$)
H2a	d_z	1.32 ($p_{\text{Holm}} < .001$)	1.51 ($p_{\text{Holm}} < .001$)	1.43 ($p_{\text{Holm}} < .001$)
H2b	d_z	1.78 ($p_{\text{Holm}} < .001$)	1.94 ($p_{\text{Holm}} < .001$)	1.85 ($p_{\text{Holm}} < .001$)
H2c	d_z	-1.32 ($p_{\text{Holm}} < .001$)	-1.65 ($p_{\text{Holm}} < .001$)	-1.39 ($p_{\text{Holm}} < .001$)
H2d	d_z	-0.88 ($p_{\text{Holm}} < .001$)	-0.88 ($p_{\text{Holm}} < .001$)	-0.87 ($p_{\text{Holm}} < .001$)
H3 arousal	r	.993 (lower .990)	.990 (lower .984)	.993 (lower .989)
H3 valence	r	.937 (lower .893)	.925 (lower .869)	.927 (lower .876)
H4a	ρ	.08 ($p_{\text{Holm}} = .355$) [†]	.03 ($p_{\text{Holm}} = .775$) [†]	.08 ($p_{\text{Holm}} = .380$) [†]
H4b	ρ	-.32 ($p_{\text{Holm}} = .001$)	-.30 ($p_{\text{Holm}} = .010$)	-.32 ($p_{\text{Holm}} = .001$)
H4c	ρ	.18 ($p_{\text{Holm}} = .098$) [†]	.21 ($p_{\text{Holm}} = .088$) [†]	.17 ($p_{\text{Holm}} = .142$) [†]
H5a	ρ	.24 ($p_{\text{Holm}} = .008$)	.24 ($p_{\text{Holm}} = .018$)	.24 ($p_{\text{Holm}} = .008$)
H5b	ρ	-.38 ($p_{\text{Holm}} < .001$)	-.32 ($p_{\text{Holm}} = .003$)	-.39 ($p_{\text{Holm}} < .001$)
H6a	ρ	.56 ($p_{\text{Holm}} < .001$)	.49 ($p_{\text{Holm}} < .001$)	.56 ($p_{\text{Holm}} < .001$)
H6b	ρ	.34 ($p_{\text{Holm}} < .001$)	.29 ($p_{\text{Holm}} = .004$)	.36 ($p_{\text{Holm}} < .001$)

Note. Cells show the effect size and, in parentheses, the Holm-adjusted p value (H1: unadjusted, single test) or, for H3, the lower bound of the 95% CI of r . [†]Hypothesis not supported under the registered criteria. Stricter engagement: sparse raters (fewer than one rating bout per minute or more than 180 s without input), participants who failed one attention check and participants who reported not watching attentively removed. Rejection set removed: participants whose Prolific submissions were rejected for behaviour on the free-text page removed, although they pass the registered exclusion rules.

wide, reflecting a gradual rise rather than a sharp change. As preregistered, all tests used the fixed edges.

Exploratory Analysis: The Affect Grid by Film Phase

The following analysis was not preregistered as a test and is purely descriptive. Figure 6 shows where on the valence–arousal grid participants spent their time in each segment. At baseline, ratings were spread around the centre of the grid with a slight lean towards low arousal. During the rise they moved upwards and to the left. On the plateau most participants converged on the high-arousal, unpleasant corner, which was by far the most occupied region of the grid at any point in the film. During the return they dispersed again around the centre, with roughly as many participants on the pleasant as on the unpleasant side.

Figure 6*Occupancy of the Affect Grid in Each Film Segment (Exploratory)*

Note. Blue shading shows the share of viewing time spent in each region of the grid (smoothed; each participant weighted equally; same colour scale in all panels). Dots are participants' mean ratings in the segment; the open circle is the group mean with 95% confidence intervals.

Exploratory Analysis: Age and Sex

Age and sex were available for 121 participants of the primary sample (59 female, 62 male). The film affected women and men, and younger and older participants, in much the same way (Table 5).

The STICSA somatic score rose by 7.17 points on average in women and 6.44 points in men. Adjusted for age, the difference (female minus male) was 0.66 [−1.77, 3.08], $p = .593$, and the change did not depend on age (per 10 years, 0.24 [−0.78, 1.25], $p = .645$; Figure 8A and B).

The time course of the ratings did not differ by sex or by age (Figures 7 and 8C and D). For arousal, neither the segment $\times$ sex interaction ($\chi^2(3) = 0.28$, $p = .964$) nor the segment $\times$ age interaction ($\chi^2(3) = 4.92$, $p = .178$) was significant, and the same was true for valence (sex: $\chi^2(3) = 0.72$, $p = .869$; age: $\chi^2(3) = 1.84$, $p = .607$). The clustered-regression check agreed in every case (Table 5). Averaged over the film, women's valence ratings were somewhat lower than men's, by −4.46 [−9.91, 0.99], $p = .108$; the difference was largest during the rise (−6.43 points [−13.19, 0.34], $p = .062$), but neither was significant.

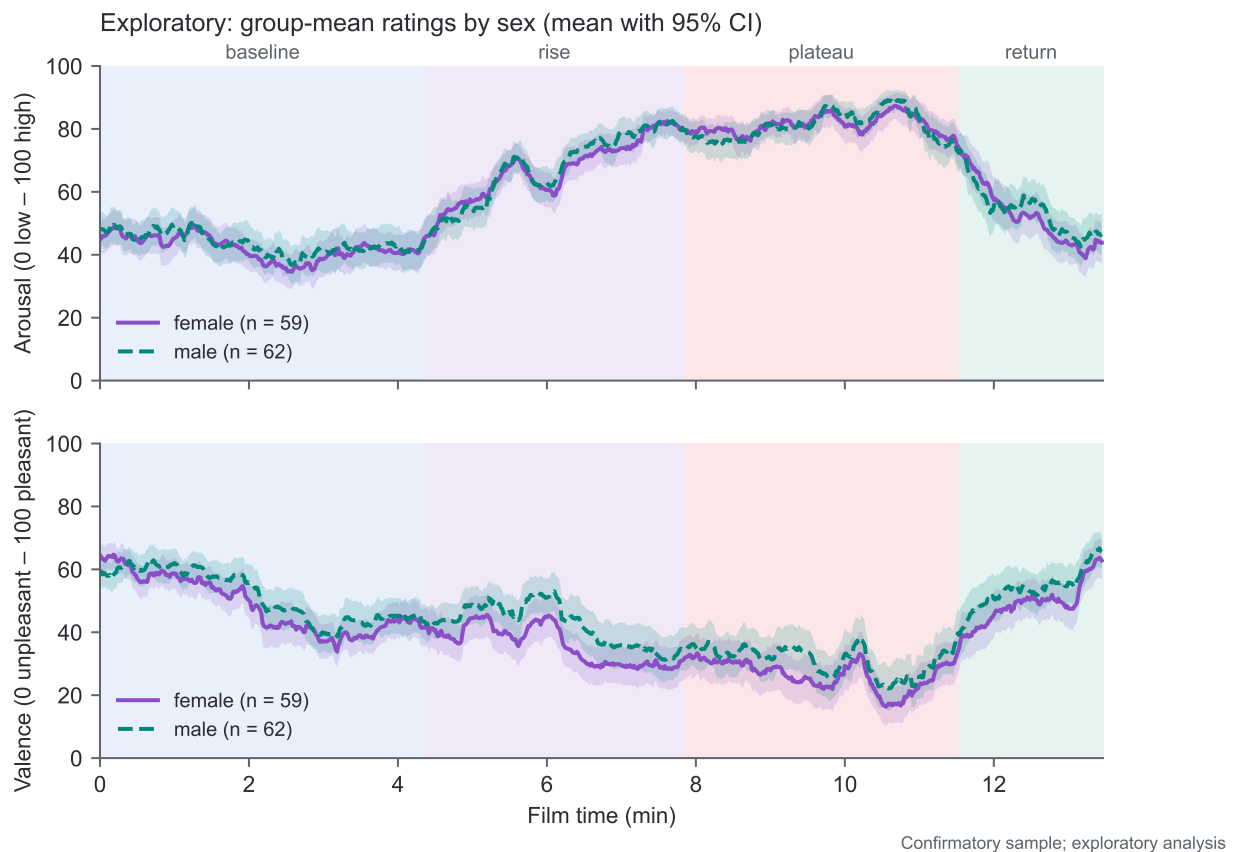
Table 5*Exploratory Analyses of Age and Sex*

Outcome	Effect	Main analysis		Clustered regression (check)	
		Test or estimate	<i>p</i>	Test or estimate	<i>p</i>
STICSA change	Female – male (points)	0.66 [–1.77, 3.08]	.593	–	–
	Per 10 years of age (points)	0.24 [–0.78, 1.25]	.645	–	–
Arousal	Segment × sex	$\chi^2(3) = 0.28$	.964	$F(3, 120) = 0.08$	.969
	Segment × age	$\chi^2(3) = 4.92$	.178	$F(3, 120) = 2.13$	.100
	Female – male (film mean)	–	–	–0.50 [–4.47, 3.47]	.806
	Per 10 years (film mean)	–	–	–1.26 [–2.88, 0.37]	.129
Valence	Segment × sex	$\chi^2(3) = 0.72$	.869	$F(3, 120) = 0.43$	.734
	Segment × age	$\chi^2(3) = 1.84$	.607	$F(3, 120) = 0.50$	.682
	Female – male (film mean)	–	–	–4.46 [–9.91, 0.99]	.108
	Per 10 years (film mean)	–	–	–0.74 [–2.96, 1.47]	.511

Note. Exploratory analyses, not preregistered tests; *p* values are uncorrected. *n* = 121 participants of the primary sample with age and sex available. STICSA change: linear regression on sex and age. Segment means: random-intercept mixed model with segment (sum-coded) × (sex + age), interactions tested with likelihood-ratio tests (main analysis); the same model fitted by least squares with standard errors clustered by participant, interactions tested with Wald *F* tests, is shown as a check, together with the main effects of sex and age averaged over the film. Age effects are per 10 years; sex effects are female minus male, in scale points.

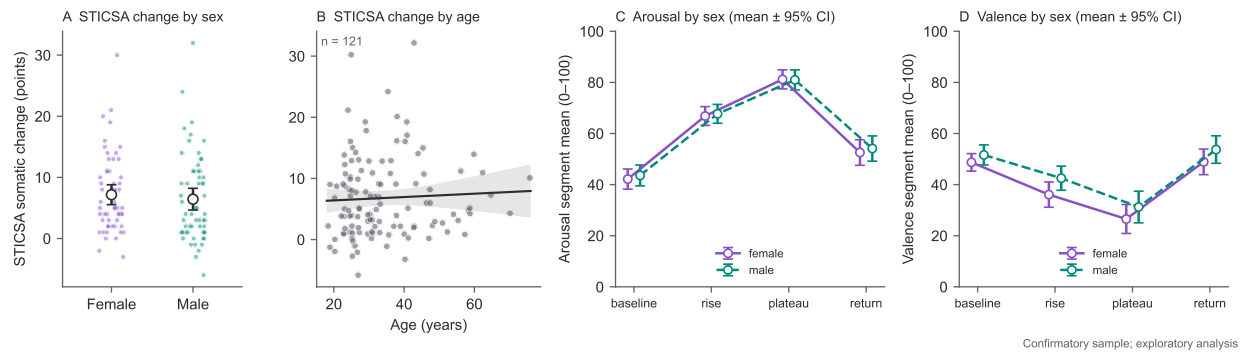
Notes on the Preregistered Plan

Five points should be read alongside the preregistration. First, the registration reported 123 usable confirmatory participants at submission; one further participant completed the study after that count was made, had complete data and passed all exclusion rules, and was included as registered (*N* = 124). Second, the fixed segment edges were estimated on 22 exploratory participants under the quality-control rules in use at the time; under the final registered rules, 21 of them are usable, and these 21 form the reference curve for H3. Third, one participant's Prolific rejection was reversed for payment reasons after the registration; the participant remains in the registered rejection set for the sensitivity analysis. Fourth, "reported not watching attentively" was implemented as answering "No" to the attentiveness question; participants who answered "Mostly" were retained. Fifth, in the exploratory age and sex analyses, the first confirmatory run reported the mixed models as not converging. This was caused by the optimization algorithm, which failed on these data while three other algorithms converged to the same solution; the code

Figure 7*Group-Mean Ratings by Sex (Exploratory)*

Note. Group means with 95% confidence intervals for female ($n = 59$, solid) and male ($n = 62$, dashed) participants. Shading marks the four preregistered segments.

was corrected to try algorithms in turn, the model and decision rule were left unchanged, and the analysis was rerun. The registration's example of mixed models over the 1-Hz series was not implemented; the models use segment means.

Figure 8*Somatic Anxiety Change and Segment Means by Sex and Age (Exploratory)*

Note. (A) STICSA somatic change by sex; points are participants, black markers means with 95% confidence intervals. (B) STICSA somatic change by age, with a least-squares line and 95% confidence band as a visual guide; points are jittered slightly because both variables take whole-number values. (C, D) Segment means of arousal and valence by sex (means with 95% confidence intervals). $n = 121$.